

FINAL EXPLANATION: BIOLOGY GRADE 11

Student Assessment Document

FINAL EXPLANATION: BIOLOGY GRADE 11: EXCRETION IN PLANTS (SUB-STRAND 2.3)

Student Assessment Document

Student Name	_____
Class	_____
Date	_____

INSTRUCTIONS FOR STUDENTS

This Final Explanation brings together everything you have learned in Lessons 1 to 5. Think about the trees and plants around our school: the Acacia and Euphorbia that ooze sticky sap or resin when cut, the droplets along the edges of grass and arrowroot (taro) leaves on a humid morning, and the old yellow leaves that fall. Answer each question in your own words in the space under it. Use the evidence you collected on the nature walks, in the microscope work, in your community interviews and portfolio, and in the class tables. Where a question asks for a Claim, Evidence and Reasoning, write all three parts. Use correct biology words such as excretion, exudation, guttation, transpiration, stomata, lenticels, hydathodes and tissue fall.

Part 1: Why plants do not need kidneys

Plants have no kidneys, bladders or sweating systems, yet they do not poison themselves with their own wastes. (a) What is excretion? (b) Explain why plants do not need complex excretory organs like the kidneys of humans. Use evidence from the metabolic rate chart and the mesophyll diagrams of CO₂ and O₂ that you studied, and from what you learned about the plant parts where wastes are kept.

Write your answer here:

Part 2: How wastes leave the plant

Describe the different ways that plants get rid of wastes. For each route, name the waste (or substance) that leaves and the part of the plant involved. Then use your observations from the nature walk and the bell jar experiment to explain what is happening to (a) the milky sap on a cut Euphorbia and the hardened gum on Acacia, and (b) the droplets at the edges of grass and arrowroot leaves in the early morning. How is guttation different from transpiration?

Write your answer here:

Part 3: Where do insoluble wastes go?

Write a Claim, Evidence and Reasoning (CER) explanation to answer this question: How do plants get rid of wastes that cannot dissolve in water or evaporate? Use what you saw when you compared young green leaves with old yellow or brown leaves, and what you saw under the microscope, including how the crystals behaved when you added dilute acid. Explain why falling leaves can be called excretion.

Write your answer here:

Part 4: Why plant wastes are valuable in Kenya

Many plant excretory products sell well in Kenyan markets. (a) Name at least four types of plant excretory products from your class table, and for each give an example of a plant or place in Kenya and one use. (b) Explain why something that is waste to the plant is valuable to people. (c) Explain what the Lesson 5 discussion taught you about the risks of some of these products.

Write your answer here:

Part 5: Final explanation and your portfolio

Write your own final explanation of the driving question: How do trees and crops clear out their metabolic wastes without any organs or excretory systems, and why are these waste products worth so much money in our markets? Your answer must use a Claim, Evidence and Reasoning structure. Use at least one piece of evidence from your community interview and photography portfolio or poster. Finish with the new question that your group has about the use of these products.

Write your answer here:

RUBRIC			
Criterion	Excellent (4)	Proficient (3)	Developing (1–2)
Explaining why plants lack excretory organs	Gives all three reasons (low metabolic rate, recycling of CO ₂ and O ₂ , storage of insoluble wastes) and links each to evidence from the chart or mesophyll diagrams.	Gives two or three reasons correctly, with limited reference to evidence.	Gives one reason or confuses excretion with other processes, with little evidence.
Describing the routes of waste removal	Correctly names and explains diffusion, transpiration, exudation, guttation and tissue fall, and clearly distinguishes guttation from transpiration using local observations.	Names most routes correctly and links them to the phenomenon, but the distinction between guttation and transpiration is partly unclear.	Names few routes or mixes them up (for example guttation and transpiration).
Use of Claim-Evidence-Reasoning with data	Claim is clear; evidence is specific (microscope, hand lens, bell jar, portfolio); reasoning links the evidence to the claim.	Claim and evidence are present, but the reasoning is brief or the link is not fully explained.	Claim is missing or unsupported, or evidence is general and not from the lessons.
Economic value and risks of plant excretory products	Gives at least four correct products with Kenyan examples and uses, explains why waste to the plant is valuable, and discusses alkaloid risks and substance abuse.	Gives two or three products with uses and a partial explanation of value or risk.	Lists one product or gives uses without explanation, with no mention of risks.

Scientific vocabulary and communication	Uses terms such as excretion, stomata, lenticels, hydathodes, exudation and guttation accurately in a clear, organised answer and ends with a relevant new question.	Uses most terms correctly and is generally clear.	Uses few scientific terms correctly and the answer is hard to follow.
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ATTRIBUTION AND LICENSING

Curriculum Alignment: Structured in pedagogical alignment with the competence-based curriculum format and learning outcomes established by the Kenya Institute of Curriculum Development (KICD). This material is independently developed and is not officially endorsed or published by KICD.

Content & Pedagogy: Developed and adapted by the educator team at SeaVuria (registered 501(c)(3) in the United States and registered NGO in Kenya, No. 872-850A-BF11), who designed the pedagogical framework, lesson templates and exemplar lessons on which this material is based.

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